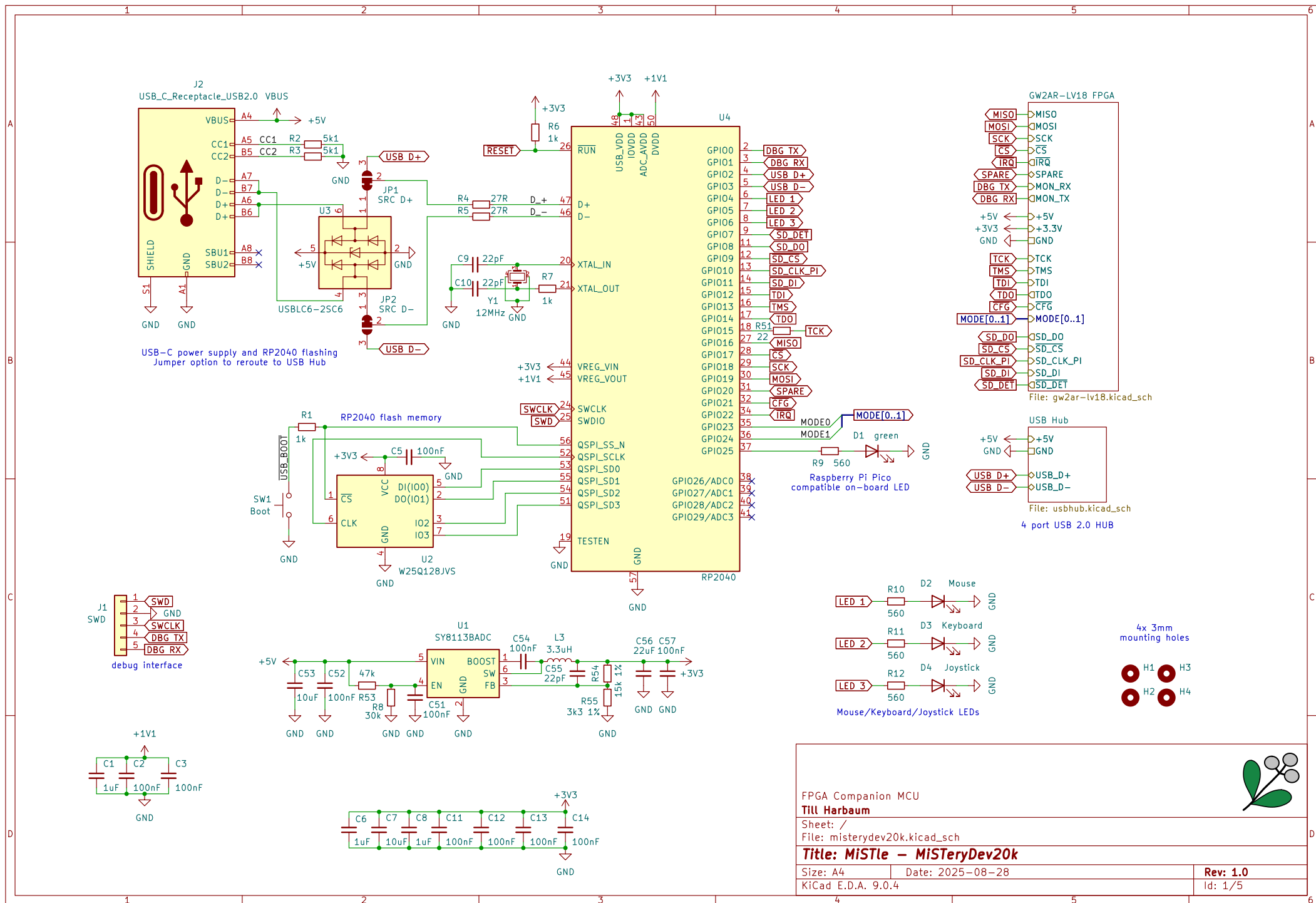
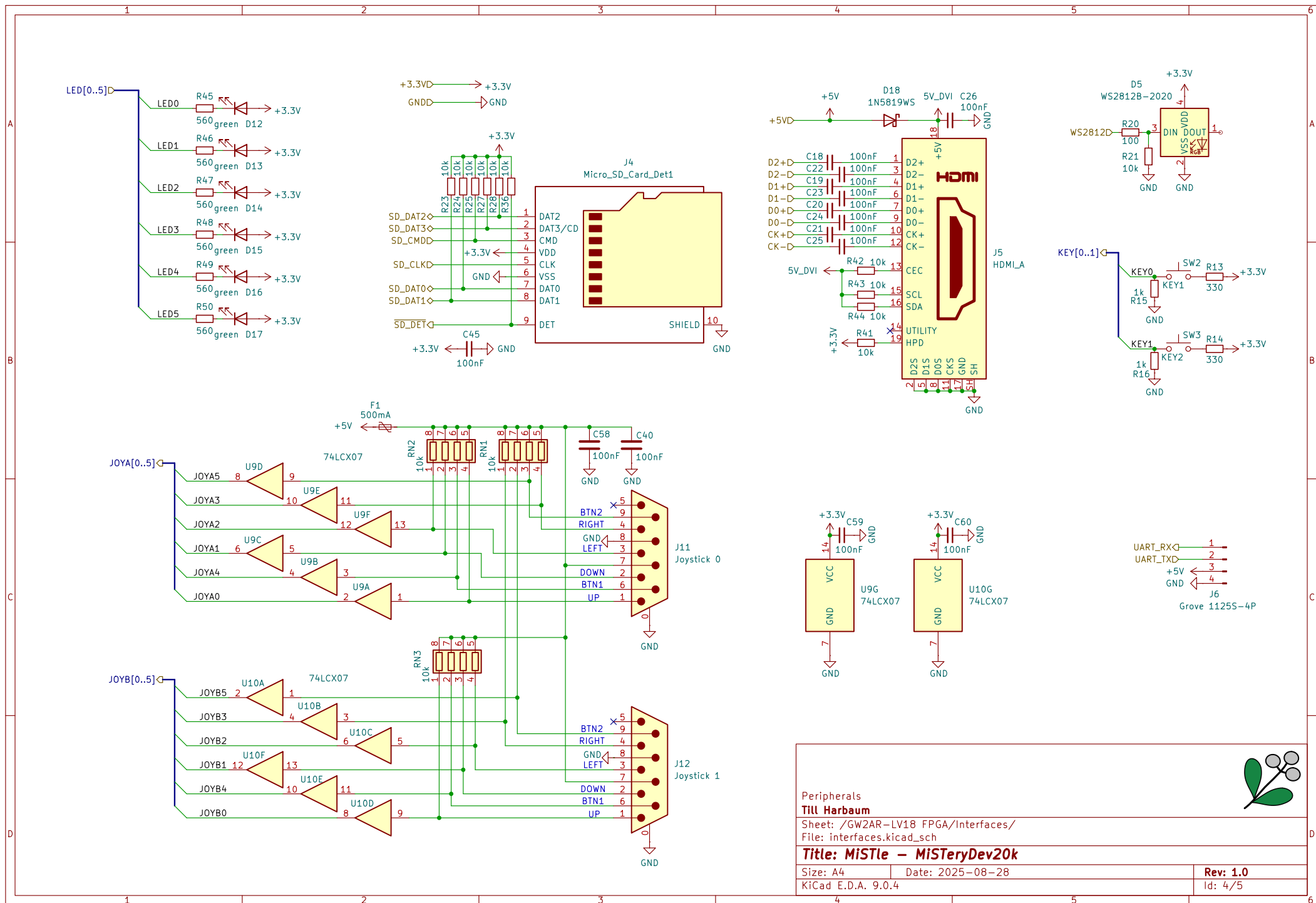






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| Pin | Function | TN20k<br>(yellow=on hdr) | NanoMig    | Pin | Function | TN20k<br>(yellow=on hdr) | NanoMig   |
|-----|----------|--------------------------|------------|-----|----------|--------------------------|-----------|
| 1   | Vcore    |                          |            | 45  | Vcore    |                          |           |
| 2   | Vss      |                          |            | 46  | Vss      |                          |           |
| 3   | Vccc     |                          |            | 47  | EXTR     |                          |           |
| 4   | IOL7A    | SYS_CLK                  | CLOCK      | 48  | IOR49B   | LCD_DE                   | MON_TX    |
| 5   | IOR25B   | SYS_TMS                  |            | 49  | IOR49A   | LCD_BL                   | SPARE4    |
| 6   | IOR26A   | SYS_TCK                  |            | 50  | Vccpll   |                          |           |
| 7   | IOR26B   | SYS_TDI                  |            | 51  | IOR45A   | PA_EN                    | MOS4      |
| 8   | IOR25A   | SYS_TDO                  |            | 52  | IOR39A   | EDID_DAT                 | SPARE5    |
| 9   | IOR31B   | RECFG                    |            | 53  | IOR38B   | EDID_CLK                 | MOS5      |
| 10  | IOL29A   | 5351_CKP                 |            | 54  | IOR38A   | I2S_DIN                  | MOS3      |
| 11  | IOL29B   | 5351_CKN                 |            | 55  | IOR36B   | I2S_LRCK                 | MON_RX    |
| 12  | Vccx     |                          |            | 56  | IOR36A   | I2S_BCLK                 | MOS2      |
| 13  | IOL45A   | HSPI_SCLK                | BL616_SCLK | 57  | IOR35A   | MSPI_WP                  | MSPI_WP   |
| 14  | Vccpll   |                          |            | 58  | Vcc      |                          |           |
| 15  | IOL47A   | LED0                     | LED0       | 59  | IOR34B   | MSPI_CLK                 | MSPI_CLK  |
| 16  | IOL27B   | LED1                     | LED1       | 60  | IOR34A   | MSPI_CS                  | MSPI_CS   |
| 17  | IOL49A   | LED2                     | LED2       | 61  | IOR33B   | MSPI_DI                  | MSPI_DI   |
| 18  | IOL49B   | LED3                     | LED3       | 62  | IOR33A   | MSPI_DO                  | MSPI_DO   |
| 19  | IOL51A   | LED4                     | LED4       | 63  | IOR29A   | MSPI_HOLD                | MSPI_HOLD |
| 20  | IOL51B   | LED5                     | LED5       | 64  | Vccx     |                          |           |
| 21  | Vss      |                          |            | 65  | Vss      |                          |           |
| 22  | Vcore    |                          |            | 66  | Vcore    |                          |           |
| 23  | Vcc      |                          |            | 67  | Vcc      |                          |           |
| 24  | Vss      |                          |            | 68  | Vss      |                          |           |
| 25  | IQB6A    | LCD_HS                   | IO2        | 69  | IOT50A   | SYS_TX                   |           |
| 26  | IQB6B    | LCD_VS                   | IO3        | 70  | IOT44B   | SYS_RX                   |           |
| 27  | IQB8A    | LCD_B7                   | IO0        | 71  | IOT44A   | HSPI_DIN0                | MIDI_OUT  |
| 28  | IQB8B    | LCD_B6                   | IO1        | 72  | IOT40B   | HSPI_DIN1                | MIDI_IN   |
| 29  | IQB14A   | LCD_B5                   | IO4        | 73  | IOT40A   | HSPI_DIN2                | SPARE0    |
| 30  | IQB14B   | LCD_B4                   | IO5        | 74  | IOT34B   | HSPI_DIN3                | SPARE1    |
| 31  | IQB18A   | LCD_B3                   | SPARE3     | 75  | IOT34A   | HSPI_DIR                 | BL616_DIR |
| 32  | IQB18B   | LCD_G7                   |            | 76  | IOT30B   | HSPI_DAT                 | BL616_DAT |
| 33  | IQB24A   | DVI_CKP                  | DVI_CKP    | 77  | IOT30A   | LCD_CLK                  | SPARE2    |
| 34  | IQB24B   | DVI_CKN                  | DVI_CKN    | 78  | Vcc      |                          |           |
| 35  | IQB30A   | DVI_D0P                  | DVI_D0P    | 79  | IOT27B   | WS2812                   | WS2812    |
| 36  | IQB30B   | DVI_D0N                  | DVI_D0N    | 80  | IOT27A   | SDIO_D2                  | SDIO_D2   |
| 37  | IQB34A   | DVI_D1P                  | DVI_D1P    | 81  | IOT17B   | SDIO_D3                  | SDIO_D3   |
| 38  | IQB34B   | DVI_D1N                  | DVI_D1N    | 82  | IOT17A   | SDIO_CMD                 | SDIO_CMD  |
| 39  | IQB40A   | DVI_D2P                  | DVI_D2P    | 83  | IOT6B    | SDIO_CLK                 | SDIO_CLK  |
| 40  | IQB40B   | DVI_D2N                  | DVI_D2N    | 84  | IOT6A    | SDIO_D0                  | SDIO_D0   |
| 41  | IQB43A   | LCD_R4                   | MOS0       | 85  | IOT4B    | SDIO_D1                  | SDIO_D1   |
| 42  | IQB42B   | LCD_R3                   | MOS1       | 86  | IOT4A    | HSPI_CS                  | BL616_CS  |
| 43  | Vss      |                          |            | 87  | IOR20B   | KEY2                     | BTN_RESET |
| 44  | Vcc      |                          |            | 88  | IOR30A   | KEY1                     | BTN_USER  |

FPGA pin usage

**Till Harbaum**

Sheet: /GW2AR-LV18 FPGA/FPGA Pin Usage/

File: gw2ar-lv18\_pins.kicad\_sch

**Title: MiSTle – MiSTeryDev20k**

Size: A4

Date: 2025-08-28

**Rev: 1.0**

KiCad E.D.A. 9.0.4

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